

MODULE 5: CYBER ETHICS & LAWS

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Introduction to Cyber Laws

- Cyber laws are laws that apply to activities conducted through computers, the internet, and electronic communication systems.
- They help regulate online behavior, resolve disputes, prevent cybercrimes, and protect users, businesses, and governments in the digital world.
- They define what is considered legal or illegal in cyberspace, covering cybercrimes, electronic transactions, data protection, and digital signatures.

Need for Cyber Laws

1. Increase in cybercrimes such as hacking, identity theft, financial fraud, phishing, and data breaches.
2. Growth of digital transactions like online banking, e-commerce, and digital payments.
3. Protection of users' privacy and personal data in digital space.
4. Legal recognition of electronic documents and contracts.
5. Data protection and privacy laws.
6. Cybersecurity guidelines for organizations.
7. Electronic signatures and authentication systems.
8. E-commerce regulations (online transactions, digital contracts, consumer protection).
9. Regulation of social media, OTT platforms, and online content.
10. Protection from online abuse, harassment, and cyberbullying.

Evolution of Cyber Laws in India

Cyber laws in India developed mainly through the **Information Technology Act, 2000** and its later amendments.

Key Milestones

- **2000:** India introduced the **IT Act 2000**, the first law providing legal recognition to electronic records, e-signatures, and cybercrime offences.
- **2008 Amendment:** Strengthened the IT Act by adding new offences like identity theft (66C), cyber fraud (66D), child pornography (67B), and data protection requirements.

- **Introduction of CERT-In:** CERT-In (Computer Emergency Response Team–India) was established to handle incidents and issue security guidelines.
- **Digital India & E-Governance Era:** Growth of Aadhaar, e-Governance, online services, eKYC, digital payments led to stronger cybersecurity guidelines.
- **Recent Developments:** New data protection rules, cybersecurity directives, and amendments to handle ransomware, cyber terrorism, OTP frauds, and online abuse.

India's cyber law framework is continuously evolving to match technological growth and emerging threats.

Scope of Cyber Regulations in Digital Age

Cyber regulations today cover a wide range of digital activities and technologies. The scope includes:

- **Cybercrimes and criminal offences** (hacking, malware, fraud, stalking, impersonation)
- **E-commerce regulations** (online transactions, digital contracts, consumer protection)
- **Digital payments and online banking rules**
- **Data protection and privacy laws**
- **Electronic signatures and authentication systems**
- **Intellectual property rights (copyright, trademarks) in digital environment**
- **Cybersecurity guidelines for organizations**
- **Regulation of social media, OTT platforms, and online content**
- **Cyber forensics and evidence handling rules**
- **Protection from online abuse, harassment, and cyberbullying**

The scope continues to expand as new technologies (AI, IoT, blockchain, cloud) introduce new risks.

E-Commerce and E-Governance

E-Commerce and E-Governance are two major digital pillars that rely on cyber laws to ensure secure, transparent, and legally valid online interactions.

E-Commerce (Electronic Commerce)

E-Commerce (Electronic Commerce) refers to the buying, selling, marketing, and exchange of goods or services using electronic systems such as the internet, mobile apps, and digital payment platforms. **Components of E-Commerce**

1. **Online Shopping Platforms:** Websites/apps for buying and selling products (e.g., Amazon, Flipkart).
2. **Electronic Payment Systems:** Credit/debit cards, UPI, net banking, wallets, payment gateways.
3. **Electronic Data Interchange (EDI):** Automated exchange of business documents between organizations.
4. **Supply Chain & Logistics Systems:** Order tracking, warehouse management, delivery networks.
5. **Online Marketing & Advertising:** SEO, email marketing, social media promotions.
6. **Customer Support Systems:** Chatbots, helpdesks, complaint portals, CRM systems.

E-commerce depends heavily on secure transactions, consumer protection, and legal recognition of digital documents.

Legal Issues in E-Commerce

E-Commerce introduces several legal challenges that must be addressed through cyber laws:

- **Fraud and identity theft** during online transactions.
- **Privacy and data protection issues** when handling customer information.
- **Disputes regarding online contracts**, returns, and warranties.
- **Payment security issues**, such as credit card fraud and phishing.
- **Intellectual property violations**, including counterfeit products and copyright misuse.
- **Jurisdiction issues**, since buyers and sellers may be in different countries.
- **Consumer rights violations** due to misleading ads or unfair terms.
- **Cyber attacks** against e-commerce websites (DDoS, hacking, ransomware).

Cyber laws help define responsibilities, penalties, and dispute resolution processes.

Digital Contracts: Digital contracts are legally valid agreements made through electronic means.

Examples include:

- Online “I Agree” clickwrap agreements
- E-mails confirming transactions
- Electronic purchase orders & invoices
- Online subscription agreements

Under the IT Act 2000, electronic records are legally recognized as contract documents.

Electronic Signatures: Electronic signatures provide authentication and validation of electronic documents. **Types include:**

- **Digital signatures (DSC):** Based on public key infrastructure (PKI), issued by Certifying Authorities.
- **e-Sign:** Aadhaar-based authentication system.
- **Simple electronic signatures:** Typed names, scanned signatures.

Legal Purpose:

- To verify **identity** of the signer
- To ensure **integrity** and **non-repudiation** of documents
- To make electronic transactions legally enforceable

E-Governance Models

E-Governance refers to the use of ICT (Information and Communication Technology) by the government to deliver services, communicate with citizens, and improve administration.

Models of E-Governance

1. **G2C (Government to Citizen):** Services like income tax filing, Aadhaar, passports, online certificates.
2. **G2B (Government to Business):** GST portal, business registrations, license renewals.
3. **G2G (Government to Government):** Inter-department data sharing, digital police networks, e-office systems.
4. **G2E (Government to Employee):** HR portals, payroll systems, digital service records.

Applications of E-Governance

- Digital payments and e-wallets
- Online grievance redressal systems
- E-courts, e-medicine, e-education
- Direct Benefit Transfers (DBT)
- Smart city projects

- Electronic voting experiments (pilot level)

E-Governance improves transparency, efficiency, and public service delivery.

Legal and Ethical Challenges in E-Governance

Legal Challenges

- **Data protection and privacy risks** (sensitive citizen data).
- **Cybersecurity threats** against government servers.
- **Authentication issues** in digital identity systems.
- **Compliance with IT Act and data laws.**
- **Digital divide** — unequal access to technology.
- **Jurisdiction issues** in cross-state electronic data sharing.

Ethical Challenges

- Misuse of citizens' personal data.
- Over-surveillance and loss of privacy.
- Bias in automated decision-making systems.
- Transparency issues in government algorithms.
- Ensuring equality in digital access for all citizens.

Certifying Authority and Controller

Under the **Information Technology Act, 2000**, digital signatures and electronic authentication are regulated through Certifying Authorities (CAs) and the Controller of Certifying Authorities (CCA). They ensure trust, security, and legality in digital communication.

3.1 Role of Certifying Authorities (CAs)

Certifying Authorities are trusted third-party organizations authorized by the government to issue **Digital Signature Certificates (DSCs)**. Their role includes:

- Verifying the identity of individuals and organizations before issuing certificates
- Generating and issuing DSCs for secure electronic communication
- Ensuring the integrity, authenticity, and security of digital signatures
- Maintaining secure PKI (Public Key Infrastructure) systems
- Revoking or suspending certificates when necessary
- Publishing Certificate Revocation Lists (CRLs) for invalid certificates

CAs act as trust providers in digital transactions.

3.2 Digital Signature Certificates (DSC)

A Digital Signature Certificate is a **digital ID** issued to a person or organization to validate identity in electronic transactions. Key points:

- Issued under the IT Act using **Public Key Infrastructure (PKI)**
- Contains public key, user identity details, expiry date, and CA details
- Provides:
 - **Authentication** (verifies user identity)
 - **Integrity** (detects data tampering)
 - **Non-repudiation** (signer cannot deny signing)

Types of DSCs

- **Class 1:** Basic identity verification
- **Class 3:** High-security verification for e-tenders, e-auctions, legal filings

(Classes are simplified after 2021; Class 2 merged into Class 3.)

3.3 Duties and Responsibilities of CAs

CAs must follow strict rules under the IT Act and CCA guidelines. Their responsibilities include:

- Verifying user identity before issuing certificates
 - Maintaining a secure repository of issued certificates
 - Publishing Certificate Revocation Lists (CRLs) regularly
 - Protecting private keys used for certificate generation
 - Ensuring secure infrastructure, hardware security modules (HSMs)
 - Keeping audit logs and submitting reports to CCA
 - Ensuring confidentiality, integrity, and reliability of digital signature processes
 - Suspending or revoking certificates when misuse or compromise is detected
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3.4 Controller of Certifying Authorities (CCA)

The Controller of Certifying Authorities is a government regulatory authority responsible for supervising and regulating all Certifying Authorities in India.

Key functions:

- License, regulate, and oversee all Certifying Authorities
- Ensure CAs follow rules under the IT Act and CCA guidelines
- Conduct audits of CAs to verify compliance and security
- Maintain the **Root Certificate Authority of India**
- Suspend or revoke licenses of CAs for non-compliance
- Issue regulations, policies, and security standards for PKI
- Act as the apex body ensuring trust and legal validity of digital signatures

CCA ensures nationwide trust in digital signing infrastructure.

3.5 Licensing and Regulations of CAs

Certifying Authorities must obtain a **license from the CCA** to operate legally.

Licensing Requirements

- Submit application with infrastructure details
- Provide security architecture, hardware details, PKI setup
- Ensure use of audited and secure systems (HSMs, servers, policies)
- Undergo checks related to personnel, premises, and processes

Regulations

CAs must comply with:

- IT Act, 2000 and IT Amendment Act, 2008
- CCA's Digital Signature Regulations
- X.509 certificate standards
- Regular security audits and vulnerability assessments
- Maintaining logs, archives, and backup records
- Protecting signing keys and certificate databases

Failure to comply can result in fines, suspension, or cancellation of license.

Offences Under the IT Act (Information Technology Act 2000 / 2008)

- The Information Technology Act, 2000 was India's first law to address cybercrimes, electronic records, digital signatures, and online transactions.
- The 2008 Amendment strengthened the Act by adding new cyber offences, data protection rules, and stricter penalties.

4.1 Overview of IT Act 2000

The IT Act 2000 was introduced to:

- Provide **legal recognition** to electronic records and digital signatures
- Facilitate **e-commerce and e-governance**
- Define **cybercrimes** and their punishments
- Establish **Certifying Authorities** for digital signatures
- Provide a legal framework for electronic communication

It was India's first comprehensive cyber law focused on regulating activities in cyberspace.

4.2 Amendments under IT Act 2008

The **IT Amendment Act 2008** was introduced to cover newly emerging cyber threats and strengthen security.

Key changes:

- Introduced new offences like **cyber terrorism (66F)**, **identity theft (66C)**, and **cyber fraud (66D)**
- Added provisions for **data protection** and **privacy**
- Recognized **electronic signatures** (broader than digital signatures)
- Increased penalties for cyber offences
- Expanded the definition of "communication device" to include mobiles, tablets, etc.
- Provided safe-harbor protection to intermediaries (like social media platforms)
- Improved guidelines for digital evidence and cyber forensics

The 2008 amendment made the Act more relevant for modern cybercrime scenarios.

4.3 Section-Wise Cyber Offences (with descriptions)

Here is a clear explanation of each offence:

1. **Unauthorized Access (Section 43 / 66):** Accessing a computer system, network, or data without permission. Includes browsing, copying data, or entering a system illegally.
2. **Data Theft (Section 43(b)):** Stealing, downloading, or copying information from a computer or device. Covers sensitive data, business information, and personal data.
3. **Identity Theft (Section 66C):** Using someone else's digital identity, password, signature, or identification information without authorization.
4. **Cyber Stalking (Section 66E / 67 / IPC sections):** Repeatedly following, contacting, or harassing someone online through messages, emails, or social media.
5. **Hacking (Section 66):** Destroying, altering, or modifying computer resources, programs, or data with malicious intent.
6. **Publishing Obscene Content (Section 67, 67A, 67B):** Publishing, transmitting, or sharing obscene, sexually explicit, or child sexual abuse material online.

7. **Tampering with Source Code (Section 65):** Deleting, altering, or hiding computer source code used by a computer system. This includes manipulating software code, logs, or configurations. These offences cover a wide range of cybercrimes recognized under the Act.
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4.4 Penalties and Legal Consequences

The IT Act prescribes strict penalties for cyber offenses, depending on the severity of the crime.

Typical Penalties Include:

- **Fines ranging from ₹1 lakh to ₹10 lakh** depending on the offence
- **Imprisonment from 3 years to life imprisonment** (for serious offences like cyber terrorism)
- **Compensation to victims** for financial or data loss
- **Seizure of computers, devices, and digital evidence**
- **Suspension or cancellation of licenses for intermediaries**

Examples

- **Unauthorized access (66):** Up to 3 years jail or fine up to ₹5 lakh
- **Identity theft (66C):** Up to 3 years jail + fine
- **Cyber fraud (66D):** Up to 3 years jail + fine
- **Obscene material (67):** Up to 5 years jail + high fines
- **Cyber terrorism (66F):** Life imprisonment

Penalties ensure accountability and discourage misuse of digital platforms.

Intellectual Property Rights (IPR) in Cyberspace

Intellectual Property Rights (IPR) are the legal rights granted to the creator or owner of works resulting from intellectual activity in the industrial, scientific, literary, or artistic fields. In **Cyberspace**, these rights deal with how creations like software, digital music, online content, and brand names are protected and enforced across global digital networks.

6.1 Copyright Issues (Digital Content)

Copyright protects original works of authorship fixed in a tangible medium of expression, such as literary, dramatic, musical, and artistic works.

- **Key Issue:** The core problem in the digital environment is the ease and speed of **reproduction and distribution**. A single digital file (music, book, film) can be copied perfectly and shared globally instantly.
 - **Digital Content:** This includes software code, e-books, website design elements, digital photographs, streamed videos, and online articles.
 - **Law's Adaptation:** Laws have had to evolve to cover temporary copies made in a computer's RAM, and the rise of digital rights management (DRM) technologies to control access and copying.
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6.2 Trademarks in Online Environment

Trademarks protect signs, designs, or expressions that distinguish the goods or services of one enterprise from those of others (e.g., brand names, logos).

- **Online Use:** Trademarks appear on websites, in search engine ads (keywords), and in domain names.
- **Key Issues:**
 - **Cybersquatting:** Registering a famous trademark as a domain name in bad faith (covered in point 6.4).

- **Keyword Advertising:** Using a competitor's trademark as a hidden keyword to trigger one's own advertisement.
 - **Initial Interest Confusion:** Customers are initially confused by a competing site's use of a similar mark, even if they realize the mistake later.
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6.3 Patents and Software Protection

Patents grant the inventor exclusive rights for a limited period to prevent others from making, using, or selling the invention.

- **Software Protection:** Software can be protected by **copyright** (for the code itself) and sometimes by **patent** (for the underlying process, method, or algorithm, especially if it provides a novel and non-obvious technical effect).
 - **Key Issue:** There is significant debate globally over what constitutes a patentable **software invention** versus an unpatentable "abstract idea" or "business method." Patents are crucial for protecting proprietary new digital technologies and business processes.
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6.4 Domain Name Disputes

A **Domain Name** is an internet address (e.g., <https://www.google.com/search?q=google.com>). Disputes arise when a party registers a domain name that is identical or confusingly similar to a pre-existing trademark.

- **Cybersquatting:** The practice of registering, trafficking in, or using a domain name with the bad-faith intent of profiting from the goodwill of someone else's trademark.
 - **Resolution:** Disputes are commonly resolved through the **Uniform Domain-Name Dispute-Resolution Policy (UDRP)**, an administrative process faster and cheaper than traditional litigation.
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6.5 Cyber Piracy and Plagiarism

These refer to the unauthorized use and distribution of copyrighted or patented digital material.

- **Cyber Piracy:** The illegal copying, downloading, or distribution of digital media (music, movies, software) or copyrighted works over the internet without permission from the rights holder. This is typically done for commercial gain or widespread public access.
 - **Plagiarism:** Presenting someone else's work or ideas as your own without proper attribution. In the cyber context, this involves copying and pasting text, code, or images from the internet into one's own work. While not always a legal violation, it is a serious academic and ethical breach.
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6.6 Enforcement Challenges in IPR Online

Enforcing IPR in the borderless environment of cyberspace is highly complex.

- **Jurisdiction:** Determining which country's laws apply when the infringer and the rights holder are in different countries, and the server is in a third country.
- **Anonymity:** The ability of infringers to hide their identity using proxies and VPNs makes identifying and locating the offender difficult.
- **Scale and Speed:** The sheer volume of content and the speed at which infringing material can spread (viral sharing) overwhelm traditional enforcement mechanisms.
- **Technological Measures:** Rights holders rely heavily on technical measures like DRM, watermarking, and automated content monitoring to fight infringement, alongside legal actions.

Network Layer Security: IPsec

- **IPsec (Internet Protocol Security)** is a suite of protocols and standards that provides **security** services at the **Network Layer (SL3\$)** of the TCP/IP model. It is essential for protecting data flows between hosts, networks, or applications.
- It protects IP packets traveling over public or untrusted networks by providing **encryption, authentication, integrity, and anti-replay security**.
- IPsec is widely used in **VPNs (Virtual Private Networks)**, site-to-site connections, remote access, and secure tunneling between networks.

Core Concepts of IPsec

IPsec is not a single protocol but a framework composed of several components designed to guarantee Confidentiality, Integrity, and Authenticity for IP packets.

1. Security Services Provided

IPsec provides the following security functions for traffic between two systems (peers):

- **Confidentiality (Encryption):** Ensures that the data contents are hidden from eavesdroppers.
- **Integrity (Hashing):** Verifies that the data has not been modified in transit.
- **Authentication:** Confirms the identity of the sender and receiver.
- **Anti-Replay Protection:** Detects and rejects duplicate packets to prevent an attacker from re-sending captured packets.

2. Key Protocols

IPsec uses two main protocols to achieve these services:

Protocol	Abbreviation	Services Provided	Description
Authentication Header	AH	Integrity, Authentication, Anti-Replay	Provides integrity and source authentication for the entire IP packet (header and data). It does not offer confidentiality (encryption).
Encapsulating Security Payload	ESP	Confidentiality, Integrity, Authentication, Anti-Replay	Provides confidentiality (encryption) of the data payload and optionally provides integrity and authentication for the payload and the ESP header.

IPsec Modes of Operation

IPsec can operate in two primary modes, affecting how the security information is applied to the IP packet:

1. Transport Mode

- **Usage:** Used for **end-to-end** communication between two hosts (e.g., host-to-host VPN).
- **Operation:** IPsec headers (**AH** or **ESP**) are inserted **between the original IP header and the upper-layer protocol data (payload)**. The original IP header remains, but the payload is protected.
- **Protection:** Primarily protects the data payload; the original IP header fields are left visible.

2. Tunnel Mode

- **Usage:** Used for **network-to-network** (e.g., site-to-site VPN) or **host-to-network** communication.
- **Operation:** The **entire original IP packet** (header and data) is encrypted or authenticated. A **new IP header** is added to the outside of the protected packet.

- **Protection:** Provides protection for the entire original packet, hiding the destination IP address of the internal network. This is the mode typically used for VPNs.
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IPsec Architecture Components

The IPsec framework relies on two other critical components to manage security relationships and keys:

1. Security Association (SA)

- An **SA** is a **simplex (one-way)** logical connection that defines the agreed-upon security parameters between two peers (e.g., the encryption algorithm, the shared secret key, and the IPsec mode).
- For two-way secure communication, **two SAs** are required: one for inbound traffic and one for outbound traffic.
- Each SA is uniquely identified by a **Security Parameter Index (SPI)**, the IP destination address, and the security protocol (AH or ESP).

2. Internet Key Exchange (IKE) Protocol

- **IKE** is used to dynamically establish and maintain the **Security Associations (SAs)** and manage the cryptographic keys required by AH and ESP.
 - **Phases:** IKE typically operates in two phases:
 - **Phase 1 (IKE SA):** Establishes a secure, authenticated channel (the IKE SA) between the two peers. This is typically done using pre-shared keys or digital certificates.
 - **Phase 2 (IPsec SA):** Uses the secure channel from Phase 1 to negotiate the parameters and generate the session keys for the actual IPsec SAs (the AH or ESP SAs) that will secure the user data.
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IPsec Applications and Use-Cases

- **Secure VPNs (Remote Access & Site-to-Site)**
 - Protecting communication over public networks
 - Secure connection between branches of companies
 - Encryption for routers, firewalls, and gateway devices
 - Protecting critical infrastructure communication (banks, government)
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9. Advantages and Limitations

Advantages

- Strong security using robust encryption
- Works at Network Layer → transparent to applications
- Provides end-to-end or site-to-site protection
- Resistant to spoofing, tampering, and replay attacks

Limitations

- Complex to configure and manage
- Can reduce network speed due to encryption overhead
- Not ideal for NAT unless NAT-Traversal (NAT-T) is used